

Earnings Call Sentiment Analysis

NLP for Data Science and AI

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Abstract

This report compares several approaches to earnings-call sentiment and studies which measures best explain the stock market reaction after the call. I first implemented interpretable bag-of-words methods based on the Harvard General Inquirer, the Loughran–McDonald (LM) finance dictionary, negation-adjusted variants, and a custom dictionary refined from the corpus itself. My initial plan was to work only with the call and presentation corpus and to avoid the Q&A section because Q&A often contains irony, sarcasm, rhetorical phrasing, and analyst wording that can distort simple word-count sentiment. After inspecting the sample, however, I decided not to hard-code that restriction. Instead, I kept three parallel corpora—presentation only, management answers in Q&A, and full-call text—so that the data could show which section contains the strongest signal.

The completed bag-of-words results are clear. Full-call sentiment dominates section-specific alternatives, and finance-oriented or corpus-adapted measures outperform the general Harvard dictionary. The strongest bag-of-words model is the custom full-call dictionary, which achieves the highest t -statistic among the dictionary methods in every return specification. I then added a FinBERT benchmark from Hugging Face using the same three transcript sections and the same regression design. The updated notebook shows that FinBERT on the full-call corpus is consistently the best FinBERT section and is highly competitive with the best bag-of-words models. It slightly outperforms the bag-of-words benchmark for CAR_{01-ff3}, but the refined custom full-call dictionary remains marginally stronger for the other three return specifications. The main conclusion is therefore nuanced: contextual modeling helps, but in this dataset a carefully tuned, finance-aware bag-of-words approach remains extremely strong and easier to interpret.

1. Introduction

The assignment asks for more than one sentiment score. It asks for a comparison between several sentiment extraction methods, their correlation, and their ability to predict the abnormal stock return that follows an earnings call. I approached the project as a progression from simple, transparent dictionary methods to a contextual transformer benchmark.

My first instinct was to exclude Q&A. The reason is practical rather than theoretical: Q&A often contains irony, sarcasm, interruptions, analyst framing, repeated questions, and short conversational fragments. Those features can mislead dictionary methods because a lexicon counts words but does not understand intent. At the same time, removing Q&A completely would also throw away real information and reduce the effective amount of text available for each call. I therefore used a more defensible design: keep three parallel corpora and let the regressions decide which one is most informative. The three corpora are presentation only, management answers in Q&A, and the full call.

This decision turned out to be useful. The final results from Part 1 and Part 2 both show that the strongest signal comes from the full-call corpus, which suggests that including the broader transcript is more informative than isolating only one section.

2. Data, Corpus Construction, and Preprocessing

The notebook combines three files from the course dataset: a call-level event file, a presentation transcript file, and a segmented Q&A file. The merge key is `file_name`. The Q&A file is sorted by call, question number, and label so that transcript order is preserved before aggregation.

Three text sections are then created for every call:

1. **Presentation:** prepared management remarks only.
2. **Q&A answers:** only management answers, excluding analyst questions.
3. **Full call:** presentation text plus the full Q&A sequence.

All text is converted to lowercase, line breaks are removed, punctuation is stripped except where apostrophes help preserve contractions, repeated spaces are collapsed, and leading or trailing spaces are removed. I then count the number of words in each section and normalize sentiment by document length. The basic tone measure is

$$\text{Tone} = \frac{\text{Positive Count} - \text{Negative Count}}{\text{Total Words}}.$$

This keeps long calls from receiving mechanically larger sentiment values only because they contain more words.

Table 1: Corpus construction summary

Item	Value
Number of calls in final merged sample	2,877
Mean words in presentation section	3,353.6
Mean words in Q&A answers section	4,077.6
Mean words in full-call section	8,950.6
Maximum words in full-call section	22,780
Main return outcomes	CAR01-ff3, CAR01-Carhart, CAR-11-ff3, CAR-11-Carhart
Regression controls	surprise, historical volatility, implied volatility, log assets

3. Sentiment Methods

3.1 Bag-of-Words Models

The first block of models is intentionally simple and interpretable.

- **Harvard General Inquirer:** a general-purpose positive/negative dictionary.
- **Loughran-McDonald:** a finance-specific dictionary that is better suited to earnings-call language.
- **Negation handling:** a short backward window flips the polarity of sentiment words after terms such as **not**, **never**, or **didn't**.
- **Custom dictionary:** a corpus-adapted lexicon built from common transcript words, finance-relevant stems, and manual cleanup of false matches.

The final dictionary sizes from the executed notebook are reported below.

Table 2: Dictionary sizes used in the analysis

Dictionary / class	Number of words
Harvard positive	1,636
Harvard negative	2,005
LM positive	347
LM negative	2,345
LM uncertainty	297
Custom positive	131
Custom negative	119

3.2 FinBERT Benchmark

To satisfy the LLM requirement, I added a contextual benchmark based on ProsusAI/finbert from Hugging Face. The FinBERT section mirrors Part 1 as closely as possible. It uses the same three corpus variants, scores each transcript in chunks because the text is too long for one pass, aggregates chunk probabilities into one transcript-level score, standardizes the resulting tone variables, and runs the same OLS regressions as the bag-of-words part.

The transcript-level FinBERT tone is defined as

$$\text{FinBERT Tone} = P(\text{positive}) - P(\text{negative}).$$

The notebook uses a simple chunking rule of 220 words per block and combines chunk-level predictions with weighted averages based on chunk length. This kept the implementation transparent and made the LLM section easy to compare with the dictionary section.

4. Results and Key Findings

4.1 Main bag-of-words result

The strongest empirical pattern in Part 1 is that **full-call sentiment performs better than section-specific sentiment**. Presentation-only and Q&A-only measures are statistically useful, but they are weaker than the best full-call measures in every return specification shown in the notebook output.

A second clear finding is that **finance-specific and corpus-adapted measures outperform the general Harvard dictionary**. Harvard and Harvard with negation remain statistically significant in some regressions, but they are consistently weaker than LM, LM with negation, and the refined custom dictionary. This is exactly what one would expect in a financial

text setting: domain-specific vocabulary matters.

The best-performing bag-of-words model in the notebook is the **custom full-call dictionary**. It has the highest t -statistic among the dictionary methods in all four reported return specifications. LM with negation and plain LM on the full-call corpus are close behind. Negation handling helps slightly for LM, but the gain is incremental rather than transformational.

Table 3: Top-performing bag-of-words specification by return outcome

Outcome	Best model	Coeff.	t -stat	Adj. R^2
CAR01-ff3	Custom, full call	0.007170	6.657	0.0465
CAR01-Carhart	Custom, full call	0.007242	6.578	0.0441
CAR-11-ff3	Custom, full call	0.007242	6.578	0.0441
CAR-11-Carhart	Custom, full call	0.007105	6.499	0.0453

Table 4: Best measure in each bag-of-words family for CAR01-ff3

Family	Section	Coeff.	t -stat	Adj. R^2
Harvard	Full call	0.004394	3.623	0.0377
Harvard + negation	Full call	0.004468	3.686	0.0379
LM	Full call	0.007251	6.201	0.0468
LM + negation	Full call	0.007315	6.251	0.0471
Custom refined	Full call	0.007170	6.657	0.0465

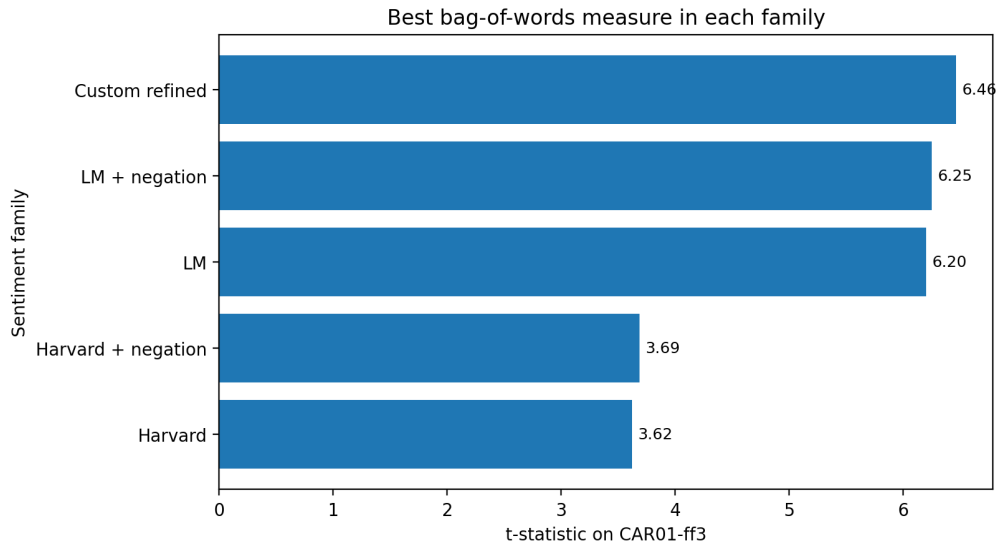


Figure 1: Comparison of the best bag-of-words measure in each sentiment family for CAR01-ff3.

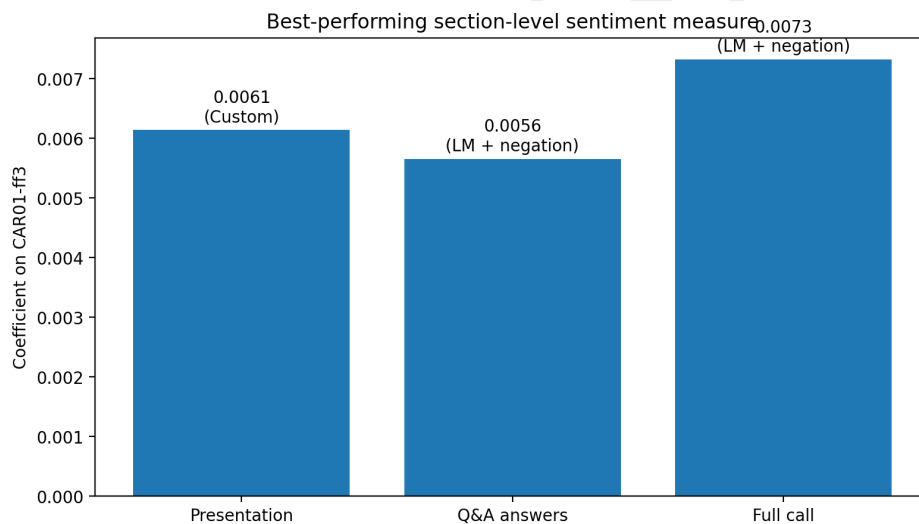


Figure 2: Best-performing section-level sentiment measure for CAR01-ff3. Full-call text is clearly strongest.

4.2 FinBERT results

The updated notebook confirms that the same section ranking also appears for FinBERT. **Full-call FinBERT is the strongest FinBERT specification in all four return outcomes**, presentation is second, and Q&A answers are weakest. This is an important consistency check because it shows that the corpus conclusion does not depend on whether sentiment is measured with a simple lexicon or with a contextual transformer.

In terms of predictive strength, FinBERT performs very well. For CAR01-ff3, the full-call

FinBERT score has coefficient 0.007409, $t = 6.776$, and adjusted $R^2 = 0.0475$, which slightly exceeds the strongest bag-of-words result. For the other three return specifications, however, the custom full-call bag-of-words model remains marginally stronger. So the contextual model is not a decisive winner in this dataset; instead it is a strong benchmark that produces results of a very similar order of magnitude.

Table 5: Best FinBERT section by return outcome

Outcome	Best FinBERT section	Coeff.	t -stat	Adj. R^2
CAR01-ff3	Full call	0.007409	6.776	0.0475
CAR01-Carhart	Full call	0.007149	6.420	0.0438
CAR-11-ff3	Full call	0.007149	6.420	0.0438
CAR-11-Carhart	Full call	0.007070	6.386	0.0452

Table 6: Direct comparison between the best bag-of-words model and full-call FinBERT

Outcome	Best bag-of-words model	Best BoW t	FinBERT t	FinBERT Adj. R^2	Stronger model
CAR01-ff3	Custom, full call	6.657	6.776	0.0475	FinBERT by a small margin
CAR01-Carhart	Custom, full call	6.578	6.420	0.0438	Custom bag-of-words
CAR-11-ff3	Custom, full call	6.578	6.420	0.0438	Custom bag-of-words
CAR-11-Carhart	Custom, full call	6.499	6.386	0.0452	Custom bag-of-words

4.3 Interpretation

The findings suggest four practical conclusions. First, using the full call is better than trying to simplify the corpus too aggressively at the start. My original concern about sarcasm and irony in Q&A was valid, but the empirical results show that, once the corpus is structured carefully, the broader transcript still carries the most predictive information. Second, dictionary choice matters more than cosmetic complexity. A general lexicon is not enough in this setting; finance-specific and corpus-adapted vocabularies perform better. Third, negation handling is useful, but only as a refinement. It improves LM slightly, yet it does not overturn the basic ranking of methods. Fourth, FinBERT is a credible contextual benchmark, but it does not dominate the simpler methods. In this sample it is best for one specification and roughly tied or slightly below the custom full-call dictionary elsewhere.

5. Conclusion

This project started from a realistic modeling concern: whether Q&A should be excluded because it contains too much conversational noise for sentiment analysis. Instead of imposing that choice too early, I kept three parallel corpora and used the data to evaluate them. That turned out to be the right decision. Both the dictionary models and FinBERT show that full-call sentiment is the most informative section across the main return measures.

The executed results also show a consistent ranking of sentiment methods. Harvard works as a baseline, but it is weaker than the finance-oriented LM dictionary and weaker still than the custom refined dictionary built from the corpus itself. Negation handling is a sensible improvement, but it is only a marginal upgrade relative to the choice of dictionary and corpus section. Among the bag-of-words measures, the strongest completed result is the custom full-call sentiment measure.

The FinBERT extension provides a useful contextual benchmark and strengthens the report because it tests whether a transformer adds value beyond word-count sentiment. The final results are mixed but informative. Full-call FinBERT is clearly the best LLM section and slightly outperforms the leading bag-of-words model for CAR01-ff3. However, the custom full-call dictionary remains slightly stronger for the other three outcomes. My final conclusion is therefore that a fine-tuned finance-domain BERT model is competitive and valid, but the simpler finance-aware bag-of-words approach remains highly effective, easier to interpret, and not clearly dominated in this dataset.

A. Appendix: Implementation notes from the notebook

- The merged sample contains 2,877 earnings calls.
- The notebook constructs presentation text, all Q&A text, question-only text, answer-only text, and full-call text before selecting the three main corpora used in the regressions.
- Bag-of-words scoring is implemented in a beginner-friendly way using explicit loops, dictionary sets, and a normalized tone formula.
- The custom dictionary is built from frequent corpus words, positive and negative seed stems, and a short manual refinement step that removes false matches.
- The FinBERT extension uses Hugging Face loading, chunk-based transcript scoring, weighted chunk aggregation, and the same regression function as Part 1.
- After recreating `log_atq` in `finbert_df`, the final FinBERT regression block runs correctly and produces a full comparison with the bag-of-words part.

B. Appendix: Final comparison summary

One concise summary of the final finding is:

“Using the same corpus design and the same controls, the contextual FinBERT benchmark was highly competitive with the strongest full-call bag-of-words measures. It slightly outperformed the leading bag-of-words model for CAR01-ff3, but the refined custom dictionary remained marginally stronger for the other return specifications. This suggests that contextual understanding adds some value, but simple finance-specific lexicons already capture most of the useful signal in this dataset.”